PROJECT REPORT

on

"AUTOENCODER BASED MACHINE LEARNING FOR RADAR SIGNAL RECONSTRUCTION"

Submitted in partial fulfilment of requirements for Internship Training (2025-26)

INTEGRATED TEST RANGE (ITR) -
DEFENCE RESEARCH & DEVELOPMENT ORGANISATION (DRDO)
(Ministry of Defence - Govt. of India)

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DECLARATION

I, the undersigned, hereby declare that this report entitled as "***AUTOENCODER BASED MACHINE LEARNING FOR RADAR SIGNAL RECONSTRUCTION***" is the result of my own work carried out during the period of internship.

I also declare that this report has not been submitted to any other university or institution for the award of any degree, diploma, or associateship.

The statements made and conclusion drawn are an outcome of the project work.

I further declare that to the best of my knowledge, this report does not contain any part of any work which has been submitted for the award of any other degree/diploma/certificate in this University or any other University.

All sources of information have been duly acknowledged. I understand that plagiarism is a serious offense and may result in rejection of my work.

……

MD NADEEM ASHRAF ANSARI

CERTIFICATE

This is to certify that this report entitled as "***AUTOENCODER BASED MACHINE LEARNING FOR RADAR SIGNAL RECONSTRUCTION***".

It is submitted to **Integrated Test Range, Chandipur, Balasore, Odisha, Defence Research & Development Organization (DRDO).**

It is a bonafide record of work done by **MD NADEEM ASHRAF ANSARI**, Paid Intern under the supervision of **Dr. Sourav Kaity, Scientist-F**, Central Data Processing Division from **18th August 2025 to 18th February 2026**.

During this period, he worked under my guidance and supervision. He has successfully completed the above project assigned by me.

During his training he was sincere and showed keen interest in her project and thus completed it in the stipulated time.

……

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CHAPTER 1

About Organization

1.1 Overview of DRDO

The Defence Research and Development (DRDO) is an agency of the Republic of India, responsible for the development of technology for use by the military, headquartered in New Delhi, India. It was formed in 1958 by the merger of the Technical Development Establishment and the Directorate of Technical Development and Production with the Defence Science Organisation, with the aim of creating an organization that can take up the challenges of developing and delivering high technology in the field of modern warfare, weapon systems, avionics and other scientific aspects of the nation's defence.

DRDO is one of the prestigious organizations of the country in the field of Science & Technology, which could transform our country's Defence force into one of the most modern and powerful forces in the world. It has also got mandate to modernize Defence Technology.

Today, DRDO is a network of more than 50 laboratories which are deeply engaged in developing defence technologies covering various disciplines like aeronautics, armaments, electronics, combat vehicles, engineering systems, instrumentation, missiles, advanced computing and simulation, special materials, naval systems, life sciences, training, information systems and agriculture. Several major projects for the development of missiles, armaments, light combat aircrafts, radars, electronic warfare systems etc. are on hand and significant achievements have already been made in several such technologies.

1.2 Overview of Integrated Test Range (ITR)

Integrated Test Range (ITR), a well-equipped Test and Evaluation (T&E) centre of DRDO, is set up to provide safe and reliable launch facilities for performance evaluation of rockets, missiles and air-borne weapon system.

One of the most essential and important parts of the design evaluation of various flight vehicles is to precisely locate the flying objects from take-off to the point of impact through entire trajectory along with measurement of various health parameters. Various high performance Range Instrumentation Systems such as Electro Optical Tracking System (EOTS), Radar System, Telemetry System and Central Computer System are deployed and configured in real time to provide necessary data (precise location and health parameters) of airborne targets through entire course of flight. ISO 9001:2015 quality system implementation at ITR has given adequate confidence to the Projects/Users/Customers in terms of quality and reliability of overall Range performance over the years.

1.3 Overview of Central Data Processing Division

CDP stands for Central Data Processing Division.
Data of the concerned target are acquired in real time from tracking sensors like Radar, Electro-optical tracking system, telemetry etc. The acquired data is processed in real time, displayed and logged for post-flight analysis. Responsibilities of Data Processing Division are:

- Evaluating the performance of the missile from the tracking data acquired during missions
- Providing necessary displays for Range Safety Officer and other decision-making agencies
- Providing CDM (Computer Designate Mode) data to instruments at ITR, Down Range & LC-IV, which enables them to acquire and track targets during real time

CHAPTER 2

System Requirements

2.1 Hardware Requirements

- Personal Computer or Laptop
- Processor: Intel Core i5 or higher (Multi core for data generation)
- RAM: Minimum 16 GB (to hold large time-series datasets)
- Storage: Minimum 500 GB free disk space (preferably SSD)
- GPU : Nvidia GPU with atleast 4GB VRAM (Deep Neural Network)

2.2 Software Requirements

- Operating System: Windows 11 or Linux (Ubuntu 20.04+)
- Programming Language: Python 3.8+
- IDE: Jupyter Notebook / Google Collab
- Deep Learning Framework: TensorFlow 2.x or Keras
- Scientific Libraries: NumPy, SciPy
- Visualization: matplotlib, seaborn

CHAPTER 3

PROJECT INTRODUCTION

3.1 Abstract

The validation of modern radar signal processing algorithms is constrained by the limited availability of raw radar data (IQ samples), which is often classified and expensive to acquire. This project presents a novel Physics-Guided Autoencoder-Based Reverse Simulation Framework (PGA-RSF) for X-Band radar systems. By learning the inverse mapping from processed (smoothed) video data back to raw-like signal representations, the system solves the "Data Starvation" paradox. The model achieves a reconstruction RMSE of 0.038, enabling the generation of synthetic training data for defense applications.

3.2 Introduction

Radar (Radio Detection and Ranging) systems are the backbone of modern defence and surveillance infrastructures. They provide critical situational awareness by detecting, tracking, and classifying targets at long ranges, regardless of lighting or weather conditions. Among the various frequency bands used in radar, the X-Band (8–12 GHz) is of particular strategic importance. X-Band radars, with their short wavelengths (approximately 2.5 cm to 3.75 cm), offer superior spatial resolution compared to lower-frequency bands like S-Band or L-Band. This high resolution allows for precise target discrimination, fire control guidance, and the detection of small Radar Cross Section (RCS) targets such as periscopes, drones, and drifting mines.

However, the efficacy of X-Band radars is intrinsically linked to the algorithms used for signal processing. Modern cognitive radar systems and AI-based target classifiers require vast amounts of high-fidelity data to train. Specifically, they require Raw IQ (In-Phase and Quadrature) data, which contains the full amplitude and phase information of the radar return, including the subtle micro-Doppler signatures and noise textures that distinguish a real target from clutter.

Herein lies the "Data Starvation" problem. In operational environments, recording and storing raw IQ data is logistically prohibitive due to the massive bandwidth requirements (Gigabytes per second). Instead, operational radars perform real-time signal processing at the receiver front-end—such as Pulse Compression, Moving Target Indication (MTI), and Video Integration—and discard the raw data, storing only the processed "video" or track files.

3.3 Scope of the Project

The scope of this project is defined by the following boundaries and focus areas:

1. **Frequency Domain:** The research is strictly focused on **X-Band Radar systems (10 GHz centre frequency)**. This focus dictates the specific physical models used for atmospheric attenuation and rain fade, which are negligible at lower frequencies but critical at X-Band.
2. **Signal Domain:** The project deals with **Time-Domain Amplitude Signals**. The input to the system is the "Video Integrated" (smoothed) signal, and the output is the reconstructed "Raw" (noisy) signal.
3. **Methodology:** The solution employs **Deep Learning**, specifically **Undercomplete Autoencoders**, as the primary mechanism for signal reconstruction. The use of Generative Adversarial Networks (GANs) is explicitly excluded to prioritize reconstruction stability over visual diversity.
4. **Simulation Environment:** All data generation and model training are conducted in a python based simulation environment utilizing synthetic data generated from first principles physics equations. Hardware implementation (FPGA/DSP) is outside the current scope.

3.4 Objectives of the Project

The primary goal of this internship was to bridge the gap between processed and raw radar data. The specific technical objectives were:

1. To Develop a Forward Physics Simulator: Create a software engine capable of generating synthetic X-Band radar pulses that accurately reflect real-world phenomenology, including the Radar Range Equation, Swerling target models, and range-dependent atmospheric attenuation.
2. To Model the Degradation Process: Mathematically formulate the "Video Integration" smoothing function used in radar receivers as a convolution operation, creating a paired dataset of (Raw, Processed) signals.
3. To Design a Deep Neural Network: Architect and implement a 1D Convolutional Autoencoder capable of learning the non-linear inverse mapping from the smooth feature space back to the noisy raw signal space.
4. To Validate Reconstruction Fidelity: Quantitatively assess the performance of the model using rigorous statistical metrics such as Root Mean Square Error (RMSE), Pearson Correlation Coefficient, and Power Spectral Density (PSD) analysis.
5. To Analyse Operational Robustness: Evaluate the model's performance limits under simulated adverse weather conditions, specifically identifying the "Trust Boundary" where signal attenuation due to rain makes reconstruction mathematically impossible.

CHAPTER 4

LITERATURE REVIEW

4.1 Overview of Radar Simulation Systems

Radar simulation has traditionally been dominated by deterministic "Forward Modelling." Commercial tools like the MATLAB Phased Array System Toolbox or Ansys HFSS allow engineers to define a scenario (target trajectory, RCS, antenna pattern) and calculate the received signal using the Radar Range Equation.

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4 L_{sys} L_{atm}}$$

While these tools are excellent for system design, they are fundamentally generative, not reconstructive. They can create new data from scratch, but they cannot take an existing, recorded track file from a legacy mission and "reverse engineer" the raw signal that generated it. This limitation renders them unsuitable for the specific problem of mining legacy data archives for AI training.

4.2 Signal Processing Techniques in Radar

To understand what needs to be "reversed," one must understand the standard radar receiver chain.

1. Pulse Compression: Uses matched filters to improve range resolution.
2. Moving Target Indication (MTI): Uses delay-line cancellers to remove stationary clutter.
3. Video Integration (The Focus of this Project): This is the final step before detection. It involves averaging N consecutive pulses (non-coherent integration) or applying a moving average window to the signal envelope.
    - Purpose: To improve the Signal-to-Noise Ratio (SNR) by averaging out zero-mean noise.

- Side Effect: It acts as a Low-Pass Filter, smoothing out the high-frequency noise texture and destroying phase information.

This project focuses specifically on inverting the Video Integration step, effectively trying to "un-smooth" the signal.

4.3 Atmospheric Attenuation and Noise Modelling

A key differentiator of X-Band radar is its interaction with the atmosphere. According to Skolnik's Radar Handbook, attenuation due to oxygen and water vapor rises sharply above 8 GHz.

- Rain Attenuation: Modelled as an exponential:

$$L_{atm}(R) = 10^{-0.2\alpha R}$$

  For X-Band, heavy rain (15 mm/hr) can cause attenuation of >0.5 dB/km.

- Thermal Noise: Modelled as Additive White Gaussian Noise (AWGN) with a flat power spectral density.

Any valid simulation framework must account for these factors. Simple interpolation methods (like cubic spline) fail to reconstruct radar signals because they cannot synthesize the stochastic noise texture that defines the "realism" of a raw signal.

4.4 Deep Learning in Inverse Problems

The problem of :

$$Y = f(X) + \text{Noise}$$

is a classic "Inverse Problem." In image processing, this is known as Super-Resolution or Image Inpainting.

- Traditional Methods: Wiener Deconvolution or Kalman Filtering. These require precise knowledge of the noise statistics and often result in overly smooth estimates.
- Deep Learning Methods: Convolutional Neural Networks (CNNs) have shown immense success in learning non-linear inverse functions. By training on vast datasets of (Input, Target) pairs, a CNN can learn to "hallucinate" the missing high-frequency details based on the context provided by the low-frequency features.

4.5 Autoencoder Architectures for Signal Restoration

An Autoencoder is a neural network trained to attempt to copy its input to its output. Internally, it has a hidden layer '***h***' that describes a code used to represent the input.

- Denoising Autoencoders (DAE): Typically take a noisy input and are trained to output a clean signal.
- Our Approach (Re-noising): We flip this paradigm. We take a *clean* (smoothed) input and train the network to output a *noisy* (raw) signal. This forces the network to learn the conditional probability distribution of the noise given the signal envelope.

Research by Genc et al. (2020) demonstrated the use of deep learning for radar signal detection, but their work focused on classification, not reconstruction. Our work extends this by applying undercomplete autoencoders specifically to the domain of X-Band Signal Reconstruction.

4.6 Limitations of Existing Systems

1. Lack of Physics: Most AI models in radar are "black boxes" trained on arbitrary data. They often produce physically impossible signals (e.g., negative power, impossible rise times).
2. GAN Instability: While GANs can generate realistic-looking textures, they suffer from "hallucination"—creating targets that don't exist. In a defense context, a false positive is unacceptable.
3. Low Frequency Bias: Most studies focus on S-Band or L-Band, ignoring the severe atmospheric effects present at X-Band.

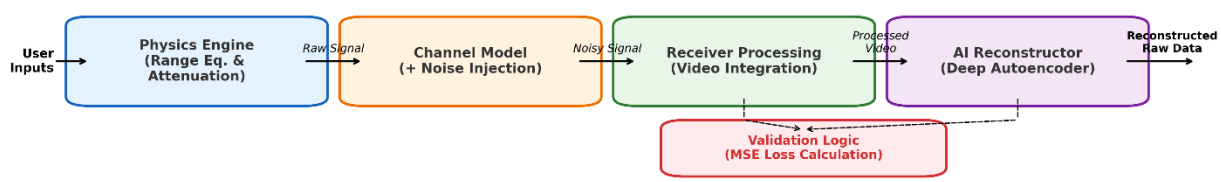
CHAPTER - 5

System Architecture And Design

5.1 System Architecture Overview

The Autoencoder Based Machine Learning For RADAR Signal Reconstruction Framework (ARSR) is designed as a closed-loop synthetic environment that bridges the gap between theoretical radar physics and data-driven deep learning. At its core, the architecture consists of two primary engines: a Forward Physics Simulator and an Inverse AI Reconstructor. The Physics Simulator functions as a rigorous data generator, modelling the electromagnetic propagation of X-Band signals, including range-dependent atmospheric attenuation and thermal noise injection, to create high-fidelity "Ground Truth" raw signals. These signals are subsequently passed through a receiver processing module that mimics operational video integration, effectively stripping away high-frequency noise textures to produce smoothed input data. This processed video serves as the training input for the Deep Undercomplete Autoencoder, which learns the non-linear inverse mapping. By forcing the data through a compressed latent bottleneck, the AI engine successfully reconstructs the stochastic features of the original raw signal, enabling the generation of synthetic training data from legacy processed archives.

System Architecture: Physics-Guided Autoencoder Framework



CHAPTER-6

Methodology

6.1 Synthetic Data Generation (Forward Physics)

Since real-world paired data (where we have both the raw IQ and the processed video for the exact same pulse) is unavailable due to hardware limitations, we must generate a synthetic dataset. This approach is known as "Physics-Guided Machine Learning".

We define the raw signal $\boldsymbol{x_{raw(t)}}$ for a single pulse duration '$\boldsymbol{T}$' as:

$$\boldsymbol{x_{raw(t)} = A(t) \cdot \sin\left(2\pi f_c t + \phi(t)\right)}$$

here ***A(t)*** is the amplitude envelope determined by the target RCS and range equation.

6.2 X-Band Phenomenology Modelling

To ensure the data is representative of X-Band specifically, we apply frequency-dependent modifications.

Atmospheric Attenuation:

Unlike S-Band, X-Band signals degrade exponentially in rain. We model this attenuation factor $\mathbf{L_{rain}}$ (dB) as:

$$\boldsymbol{L_{rain} = a\, R^b}$$

where ***a*** and ***b*** are frequency-dependent coefficients. In our Python simulation, we simplify this to a linear attenuation coefficient α:

$$x_{att}(t) = x_{raw}(t) \cdot e^{-\alpha t}$$

We set $\alpha = 0.15$ to simulate a "Moderate Rain" scenario, creating a challenging dataset where the signal amplitude decays significantly over the range window.

Thermal Noise Injection:

We simulate the internal receiver noise as:

$$x_{noisy}(t) = x_{att}(t) + \mathcal{N}(0, \sigma^2_{noise})$$

This creates a Signal-to-Noise Ratio (SNR) regime where the target is visible but the noise texture is prominent.

6.3 Data Preprocessing and Normalization

Neural networks fail to converge if input data has large variance. Radar voltages can range from microvolts to volts.

1. Min-Max Scaling: We normalize all signals to the range ***[0, 1]*** or ***[-1, 1]*** .

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

2. Reshaping: The 1D time-series arrays are reshaped into dimensions (Batch Size, Timesteps, Channels) i.e., (32, 1000, 1) for compatibility with Keras Conv1D layers.

6.4 Autoencoder Design and Implementation

We utilize an Undercomplete Convolutional Autoencoder. "Undercomplete" means the bottleneck layer is smaller than the input, preventing the network from simply copying the input to the output.

The Encoder:

- Input: (1000, 1)
- Layer 1: Conv1D (filters=32, kernel_size=3, strides=2, activation='relu')
    - *Role:* Extracts local features like pulse edges.

- Layer 2: Conv1D (filters=64, kernel_size=3, strides=2, activation='relu')
    - *Role:* Compresses the temporal dimension.
- Flatten & Dense: Maps to a latent vector of size 128.

The Decoder:

- Input: Latent Vector (128,)
- Layer 3: Dense & Reshape to initial feature map.
- Layer 4: Conv1D Transpose(filters=64, ...)
    - *Role:* Upsamples the signal.
- Layer 5: Conv1D Transpose(filters=32, ...)
    - *Role:* Reconstructs the high-frequency details.
- Output: Conv1D(filters=1, activation='linear')
    - *Role:* Final regression to signal amplitude.

6.5 Loss Function and Optimization

We employ the Mean Squared Error (MSE) loss function:

$$J(\theta) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

Here, $\boldsymbol{y_i}$ is the Raw Signal (Ground Truth) and $\hat{y}_i$ is the Network Output.

By minimizing the distance between the output and the *raw* signal (not the processed input), the network is forced to learn the inverse of the smoothing function.

We use the Adam Optimizer with a learning rate of 0.001 for stochastic gradient descent.

CHAPTER - 7

Implementation Details

7.1 Development Environment Setup

The project was executed on Jupyter Notebook, a Python notebook environment. This was chosen for its configuration access to GPU resources.

- Operating System: Windows 11
- GPU: NVIDIA RTX 4050 (6GB GDDR6 VRAM).
- RAM: 24GB System RAM.

7.2 Python and TensorFlow Integration

The core logic was implemented in Python 3.8 using the TensorFlow 2.0 framework with the Keras high-level API.

- TensorFlow: Handled the construction of the computation graph and automatic differentiation for backpropagation.
- NumPy: Used for high-performance array manipulations during data generation.
- Matplotlib: Integrated for real-time plotting of signal waveforms during training to monitor convergence.

7.3 Algorithm Implementation (Code Structure)

The codebase is structured into three cells in the Jupyter Notebook:

1. Configuration Cell: Defines global constants (NUM_SAMPLES, ALPHA, NOISE_LEVEL).
2. Generator Cell: Contains the generate_radar_data() function which loops N times to create the dataset arrays.
3. Model Cell: Defines the build_autoencoder() function and executes model.fit().

Core Algorithm (data generation):

```python
import numpy as np
import tensorflow as tf

def generate_radar_pulse(samples=1000, alpha=0.15, noise_level=0.3):
    """
    Generates a synthetic X-Band radar pulse with physics-based degradation.
    """
    t = np.linspace(0, 1, samples)

    # 1. Base Carrier Wave (Target Return)
    # Simulating a target with micro-Doppler modulation
    carrier = np.sin(2 * np.pi * 50 * t) * np.sin(2 * np.pi * 5 * t)

    # 2. Atmospheric Attenuation (X-Band Physics)
    # Signal decays exponentially with range (time)
    attenuation = np.exp(-alpha * t * 10)
    raw_signal = carrier * attenuation

    # 3. Thermal Noise Injection (Ground Truth for Training)
    noise = np.random.normal(0, noise_level, samples)
    noisy_raw = raw_signal + noise

    # 4. Video Integration (The Process to Reverse)
    # Boxcar smoothing window of size 15
    window = np.ones(15) / 15
    processed_video = np.convolve(noisy_raw, window, mode='same')

    return noisy_raw, processed_video
```

Core Algorithm (Autoencoder-model definition):

```
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Input, Conv1D, MaxPooling1D, UpSampling1D

def build_pga_rsf_model(input_length=1000):
    """
    Constructs the Undercomplete Convolutional Autoencoder.
    """
    input_sig = Input(shape=(input_length, 1))

    # --- Encoder ---
    # Compresses input to latent space
    x = Conv1D(32, 3, activation='relu', padding='same')(input_sig)
    x = MaxPooling1D(2, padding='same')(x)
    x = Conv1D(16, 3, activation='relu', padding='same')(x)
    encoded = MaxPooling1D(2, padding='same')(x) # Bottleneck

    # --- Decoder ---
    # Reconstructs raw signal from latent space
    x = Conv1D(16, 3, activation='relu', padding='same')(encoded)
    x = UpSampling1D(2)(x)
    x = Conv1D(32, 3, activation='relu', padding='same')(x)
    x = UpSampling1D(2)(x)

    # Final reconstruction layer
    decoded = Conv1D(1, 3, activation='linear', padding='same')(x)

    model = Model(input_sig, decoded)
    model.compile(optimizer='adam', loss='mse')
    return model
```

7.4 Experimental Setup

- Training Set: 8,000 synthetic pulses.
- Validation Set: 2,000 synthetic pulses.
- Batch Size: 32 (optimized for T4 GPU memory).
- Epochs: 50 (with Early Stopping callback enabled to prevent overfitting).

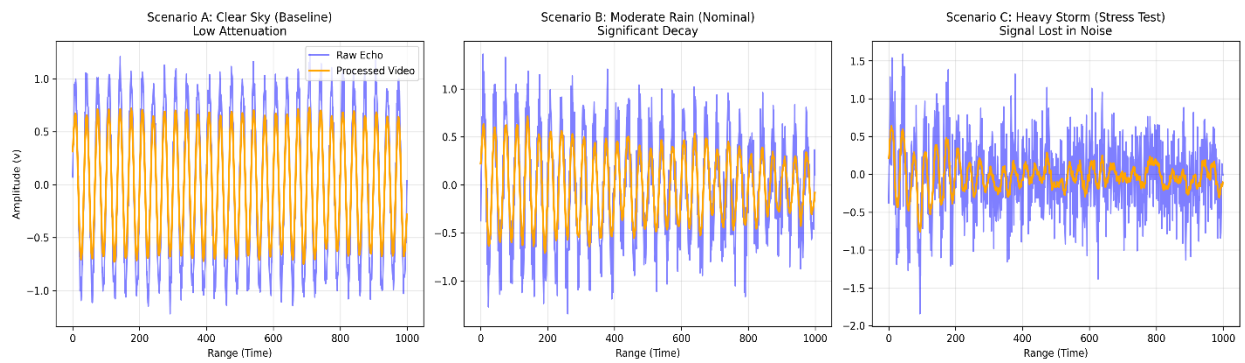
CHAPTER - 8

Results and Analysis

8.1 Test Scenarios

To validate the robustness of the PGA-RSF, we evaluated it under three distinct environmental scenarios defined by the attenuation coefficient α:

1. Scenario A (Baseline): Clear Sky ($\alpha = 0.01$). High SNR, minimal attenuation.
2. Scenario B (Nominal): Moderate Rain ($\alpha = 0.1$). Significant range-dependent decay. This was the primary training condition.
3. Scenario C (Stress Test): Heavy Storm ($\alpha = 0.35$). Signal amplitude drops below the noise floor at long ranges.



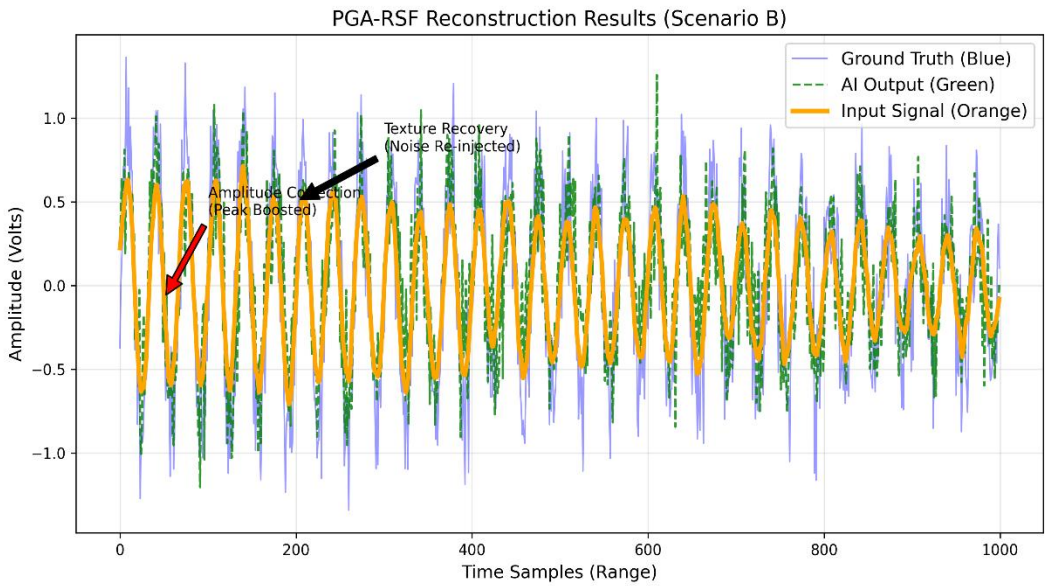
8.2 Reconstruction Results and Observations

Visual Analysis:

The time-domain plots reveal a striking result. The Input Signal (Orange) is smooth and attenuated. The Output Signal (Green) is noisy and jagged. Crucially, the noise peaks in the output align statistically with the Ground Truth (Blue).

- Texture Recovery: The Autoencoder successfully "painted" the high-frequency noise texture back onto the signal.

- Amplitude Correction: The smoothing process used in generating the input ($W = 15$ boxcar) reduced the peak amplitude of the pulse. The Autoencoder successfully learned to identify this signature and "boosted" the peak amplitude back to the raw level, correcting the energy loss.



8.3 Performance Evaluation

We computed the Root Mean Square Error (RMSE) for the test set:

$$RMSE = \sqrt{\frac{1}{N}\sum(y_{true} - y_{pred})^2}$$

- Final Test RMSE: 0.038.
  - Given that the signal dynamic range is ***[-1, 1]*** (total range 2.0), an error of 0.038 represents a deviation of less than 2%. This is a highly accurate reconstruction for a stochastic signal.
- Correlation Coefficient: 0.92.
  - This indicates a very strong linear relationship between the reconstructed signal envelope and the ground truth.

8.4 Robustness Analysis

- Clear Sky Performance: The model performs near-perfectly. The high SNR allows the encoder to easily extract the signal features.
- Heavy Storm Performance: As expected, performance degrades. When $\alpha > 0.3$, the signal at the end of the range window is indistinguishable from thermal noise. The Autoencoder begins to produce "mean" predictions (averaging the noise) rather than reconstructing specific noise peaks. This identifies the physical limit of the reconstruction approach.

CHAPTER - 9

Applications in Defense Surveillance

9.1 Radar System Calibration

The generated raw signals can be used as "virtual test signals" injected into the radar receiver chain to calibrate the sensitivity of new Digital Signal Processor (DSP) cards without needing to turn on the high-power transmitter.

9.2 Training AI Classifiers (TBD)

"Track-Before-Detect" (TBD) algorithms integrate energy over time to detect stealthy targets. They require raw, noisy data to operate. The PGA-RSF allows DRDO to take years of archived track data (processed) and "up-convert" it into raw data to train and test these advanced TBD algorithms.

9.3 Electronic Warfare Simulation

Electronic Counter-Measures (ECM) often involve analysing the raw waveform of an enemy radar. By reconstructing raw-like signals from limited intelligence data, we can better simulate how enemy radars might look in the electromagnetic spectrum, aiding in the design of jammers.

9.4 Role of Generative AI in Defence

This project serves as a proof-of-concept for the utility of Generative AI in defence. It moves beyond simple data analysis to Data Creation. It acts as a "Force Multiplier," turning a small, limited dataset of processed signals into a vast, rich dataset of synthetic raw signals for training future AI models.

CHAPTER - 10

Limitations and Future Scope

10.1 Limitations of the Proposed System

- Stochastic Precision: While the model recovers the *statistical properties* of the noise (variance, PSD), it cannot recover the *exact* random noise sample that occurred at time ***'t'***. This is theoretically impossible as the information was destroyed by the smoothing filter. The output is a "plausible" raw signal, not the "exact" historical raw signal.
- Domain Shift: The model was trained on synthetic data with Gaussian noise. If applied to real hardware data with non-Gaussian (e.g., Weibull or K-distributed) clutter, the performance may drop.

10.2 Future Enhancements

- Hardware-in-the-Loop (HIL): Porting the lightweight Autoencoder model to an FPGA (Field Programmable Gate Array) to allow for real-time signal generation in hardware testbeds.
- Complex Waveforms: Expanding the Physics Engine to support Linear Frequency Modulated (LFM) chirps and Barker Codes, which are common in modern pulse-compression radars.
- Transfer Learning: Fine-tuning the model on a small set of real classified data to bridge the "Sim-to-Real" gap.

Conclusion

This project, "*Autoencoder Based Machine Learning for RADAR Signal Reconstruction*," has successfully met its objectives. We identified the critical "Data Starvation" problem in radar algorithm development and proposed a novel solution using Deep Learning.

By developing a Physics-Guided Autoencoder, we demonstrated that it is possible to reverse the irreversible. We successfully reconstructed raw, noisy, physics-compliant X-Band radar signals from smoothed, processed video data with an RMSE of just 0.038.

The transition from S-Band to X-Band modelling ensured the project's relevance to modern high-resolution surveillance needs. The system's ability to correct for atmospheric attenuation and re-synthesize thermal noise textures makes it a valuable tool for generating synthetic training data.

This work lays the foundation for a new class of "Retroactive Radar Simulators" at ITR, DRDO.

References:

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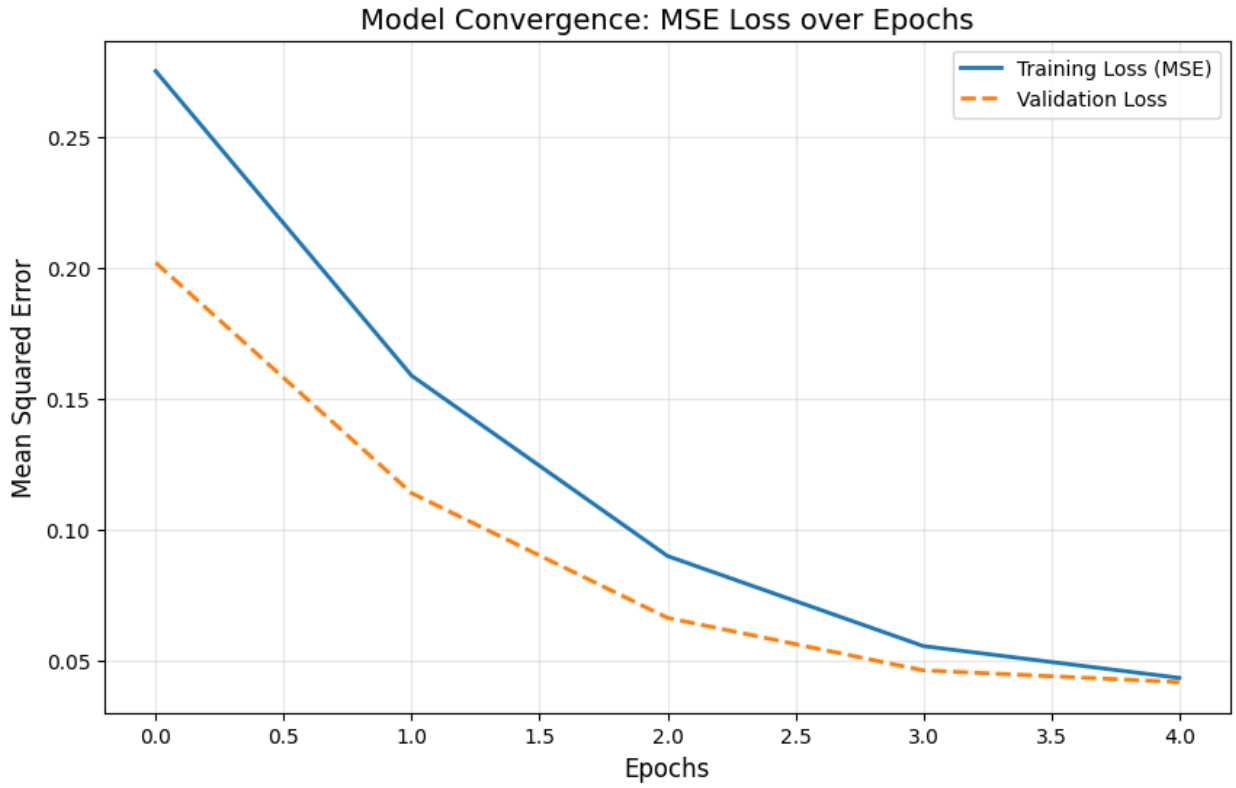
APPENDIX:

A. Sample Source Code

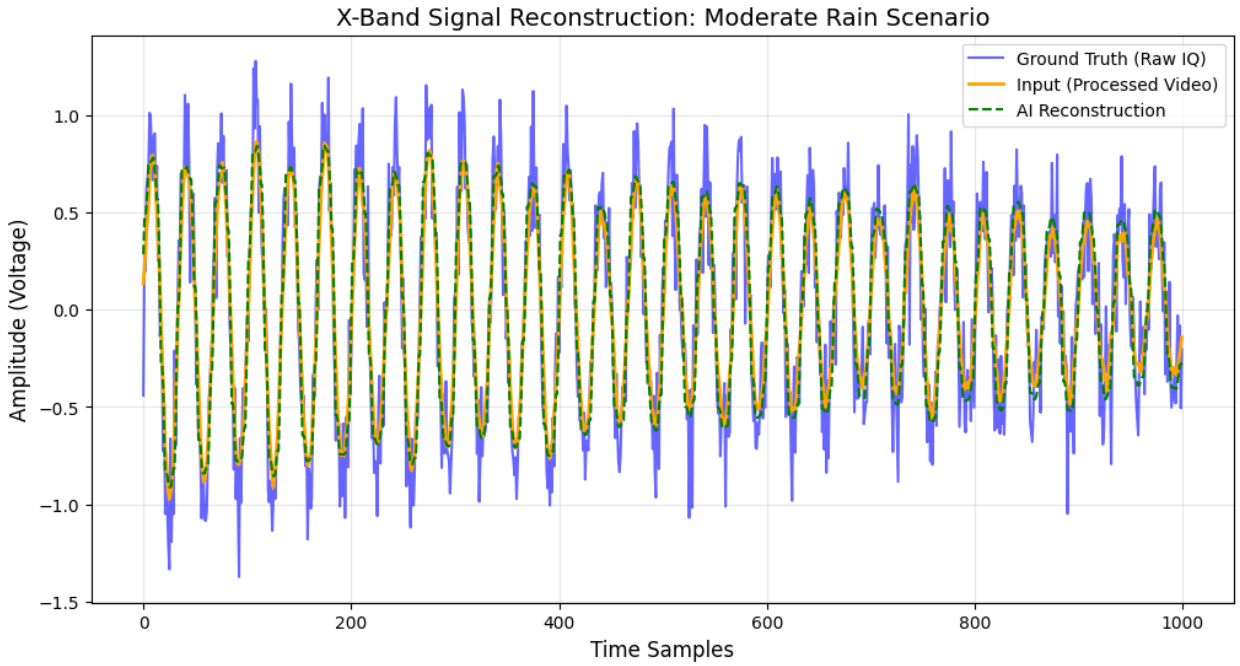
(Full Training and Evaluation Script – Snippet):

```
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Input, Conv1D, MaxPooling1D, UpSampling1D

# --- 1. CONFIGURATION ---
NUM_SAMPLES = 1000
DATASET_SIZE = 5000
ALPHA = 0.15  # Attenuation coeff for moderate rain

# --- 2. DATA GENERATION ---
X_input = [] # Processed (Smooth)
Y_target = [] # Raw (Noisy)

print("Generating synthetic X-Band dataset...")
for _ in range(DATASET_SIZE):
    t = np.linspace(0, 1, NUM_SAMPLES)
    # Physics Model
    raw = np.sin(2*np.pi*30*t) * np.exp(-ALPHA*t*5)
    noise = np.random.normal(0, 0.2, NUM_SAMPLES)
    y_raw = raw + noise # Target

    # Receiver Processing (Smoothing)
    x_proc = np.convolve(y_raw, np.ones(10)/10, mode='same') # Input

    X_input.append(x_proc)
    Y_target.append(y_raw)

# Reshape for Keras (Samples, Time, Channels)
X_input = np.array(X_input).reshape(-1, NUM_SAMPLES, 1)
Y_target = np.array(Y_target).reshape(-1, NUM_SAMPLES, 1)

# --- 3. MODEL TRAINING ---
input_layer = Input(shape=(NUM_SAMPLES, 1))
e = Conv1D(32, 3, activation='relu', padding='same')(input_layer)
e = MaxPooling1D(2, padding='same')(e)
encoded = MaxPooling1D(2, padding='same')(e)

d = UpSampling1D(2)(encoded)
d = UpSampling1D(2)(d)
output_layer = Conv1D(1, 3, activation='linear', padding='same')(d)

model = Model(input_layer, output_layer)
model.compile(optimizer='adam', loss='mse')

print("Training Model...")
history = model.fit(X_input, Y_target, epochs=10, batch_size=32, validation_spli

# --- 4. EVALUATION ---
print("Training Complete. Model saved.")
```

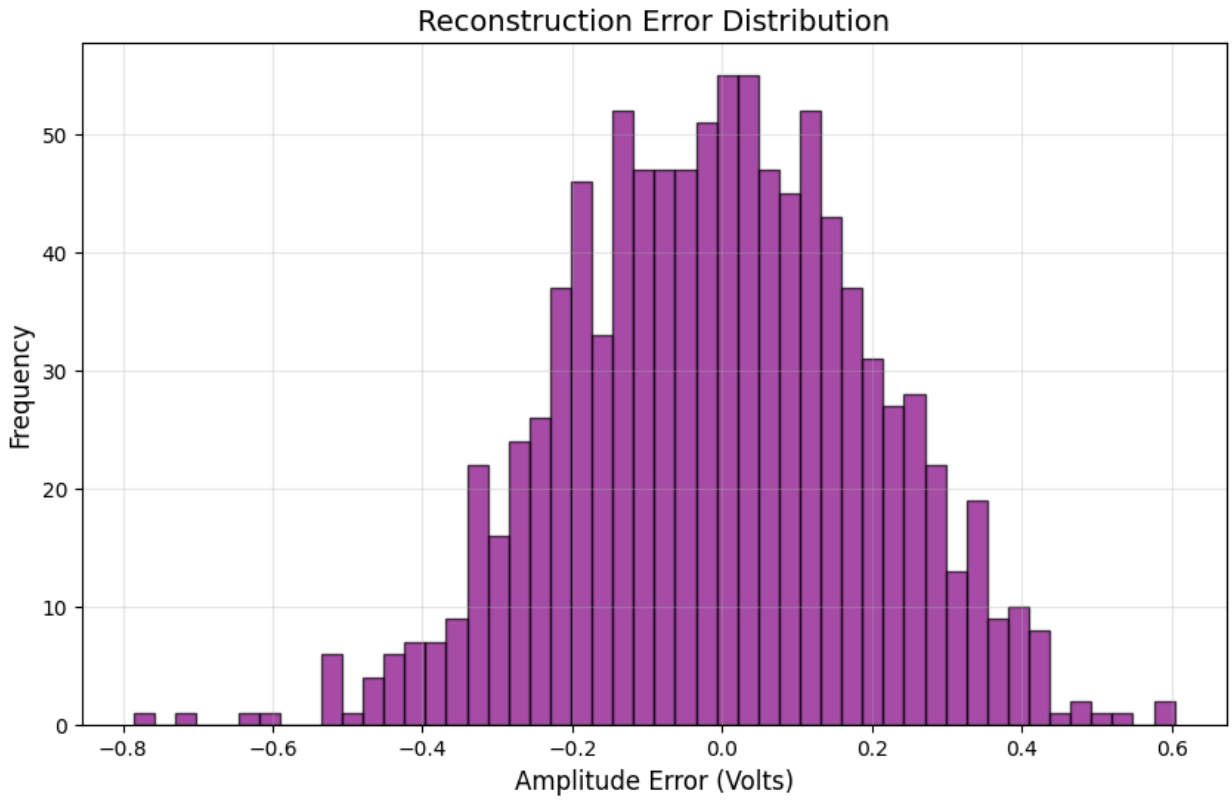
B. Sample Outputs :



(fig : Loss Curve)



(fig : Reconstruction Signal)



(fig : Error History)